A uniform logarithmic lower bound for cyclovariety Mahler measures

Sam Vaseghi*
Wisehouse Science Communication Lab, Wisehouse AB

28 October 2026

Abstract

Let C_k be the cyclovariety associated with all k-th roots of unity in the construction of Cornelissen, Hokken and Ringeling. We prove, for every integer $k \geq 2$,

$$\mathrm{m}(C_k) \geq \frac{1}{6}\log k - \frac{1}{4} + \frac{1}{3k} - \frac{1}{12k^2}.$$

Together with the known upper bound, this gives $\mathrm{m}(C_k) \asymp \log k$ uniformly in k, and proves their Conjecture 9.2.1 for every odd integer $n > 1$. The proof combines Jensen's formula, an elementary count of balanced quadruples of roots of unity, and the classical second-moment tail inequality. Only the second and fourth moments of the associated Laurent polynomial are needed; no independence among its monomials is assumed.

1 The cyclovarieties and the lower bound

For a nonzero Laurent polynomial P in r variables, write

$$\mathrm{m}(P) = \int_{\mathbb{T}^r} \log|P(x)|\, d\mu(x),$$

where $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ and μ is normalized Haar measure. All logarithms are natural. We use $\log^+ t = \max\{0, \log t\}$ for $t > 0$, with $\log^+ 0 = 0$.

Cornelissen, Hokken and Ringeling [1] introduced a family of Laurent hypersurfaces whose Mahler measures describe limits arising from Gaussian periods. Their construction retains the additive relations among roots of unity as relations among integer exponent vectors. To specify it, fix $k \geq 2$, put $d = \varphi(k)$, and let $\zeta_k = e^{2\pi i/k}$. For $0 \leq j < k$, define $a_j = (a_{j,0}, \ldots, a_{j,d-1}) \in \mathbb{Z}^d$ by

$$\zeta_k^j = \sum_{\ell=0}^{d-1} a_{j,\ell}\zeta_k^\ell. \tag{1.1}$$

These coordinates are integral because reduction modulo the monic cyclotomic polynomial has integer coefficients, and they are unique because $1, \zeta_k, \ldots, \zeta_k^{d-1}$ form a $\mathbb{Q}$-basis of $\mathbb{Q}(\zeta_k)$. For $x = (x_1, \ldots, x_d)$, set

$$F_k(x) = \sum_{j=0}^{k-1} x^{a_j}, \qquad x^{a_j} = \prod_{\ell=0}^{d-1} x_{\ell+1}^{a_{j,\ell}}, \qquad P_k(x_0, x) = x_0 + F_k(x). \tag{1.2}$$

*Email: sam.vaseghi@wisehouse-publishing.com. This version revises the exposition of the preprint dated 4 October 2026 (doi:10.5281/zenodo.23138518); the mathematical results are unchanged.

The cyclovariety is $C_k = \{P_k = 0\} \subset (\mathbb{C}^\times)^{d+1}$, and $\mathrm{m}(C_k)$ denotes $\mathrm{m}(P_k)$, as in [1, Definition A].

The known estimates include a positive constant lower bound, the upper bound $\mathrm{m}(C_k) \le \frac{1}{2}\log(k+1)$, and sharper asymptotics for prime k, powers of two, and $k = 2q^r$ with fixed r as the prime q tends to infinity [1, Propositions 6.1.1–6.1.2 and Theorem E]. The general logarithmic-order question also appears in Hokken's thesis [2, Conjecture 6.8.4]. The following estimate covers every integer $k \ge 2$.

Theorem 1.1. *For every integer $k \ge 2$,*

$$\frac{1}{6}\log k - \frac{1}{4} + \frac{1}{3k} - \frac{1}{12k^2} \;\le\; \mathrm{m}(C_k) \;\le\; \frac{1}{2}\log(k+1). \tag{1.3}$$

In particular, with

$$c_0 = \frac{(1-2^{-1/2})^2}{12} > 0,$$

one has

$$c_0 \log k \le \mathrm{m}(C_k) \le \frac{1}{2}\log(k+1) \qquad (k \ge 2). \tag{1.4}$$

Thus $\mathrm{m}(C_k) \asymp \log k$, with absolute comparison constants.

For the lower bound, the extra coordinate x_0 changes what must be controlled. Jensen's formula turns $\mathrm{m}(P_k)$ into the average of $\log^+|F_k|$. This integrand is nonnegative, so small values of $|F_k|$ cannot cancel the contribution from large values.

For Haar-distributed X, we will show that $\mathbb{E}|F_k(X)|^2 = k$ and $\mathbb{E}|F_k(X)|^4 \le 3k^2$. The second moment alone could be carried by rare, very large values; the fourth-moment bound prevents such concentration. Together, these estimates give a lower bound independent of k for the probability that $|F_k(X)| > k^{1/4}$. On this event, $\log^+|F_k(X)| \ge \frac{1}{4}\log k$, which yields the uniform logarithmic lower bound. Integrating the same tail estimate supplies the sharper expression in (1.3).

Orthogonality reduces the two moments to counts of additive relations among roots of unity. The even fourth-moment count is already given by Labelle and Lacasse [3, p. 606], and the tail estimate is the classical second-moment inequality, in the form recorded by Salem and Zygmund [4, Lemma (4.2.4)]. We give the short argument, including the balanced count for odd k, to show how these established ingredients yield the uniform cyclovariety conclusion.

2 The positive logarithmic integral and its moments

We first integrate in the extra variable x_0, keeping x fixed. At this stage $F_k(x)$ is a single complex number w. The one-variable Jensen formula gives

$$\int_{\mathbb{T}} \log|z+w|\,d\mu(z) = \log^+|w|.$$

The identity also holds when $|w| = 1$: the possible zero on the circle has an integrable logarithmic singularity. The general extra-variable identity is recorded in Smyth [5, (7)]; its cyclovariety form is used in [1, Lemma 6.2.3].

For this polynomial, the iterated integral can be justified directly. Write $\log^- t = \max\{0, -\log t\}$ for $t > 0$, allowing the value $+\infty$ at zero. Since $|F_k(x)| \le k$, the positive part of $\log|x_0 + F_k(x)|$ is at most $\log(k+1)$. For each fixed x, the linear Jensen identity gives

$$\int_{\mathbb{T}} \log^-|x_0 + F_k(x)|\,d\mu(x_0) = \int_{\mathbb{T}} \log^+|x_0 + F_k(x)|\,d\mu(x_0) - \log^+|F_k(x)| \le \log(k+1).$$

Tonelli's theorem therefore bounds the integrals of both nonnegative parts over the full torus. Fubini's theorem applies to their difference, so for Haar-distributed $X \in \mathbb{T}^d$ we obtain

$$\mathrm{m}(C_k) = \int_{\mathbb{T}^d} \log^+ |F_k(x)|\, d\mu(x) = \frac{1}{2}\mathbb{E}\log^+ Y, \qquad Y = |F_k(X)|^2. \tag{2.1}$$

The equality includes $F_k = 0$ under our convention for $\log^+ 0$. It requires no probability density for Y.

We now bound this nonnegative logarithmic average using two moments. One threshold will suffice for logarithmic growth; integrating over thresholds gives the explicit estimate below.

Lemma 2.1. *Let Y be a nonnegative random variable on a probability space, and suppose $\mathbb{E}Y = a > 1$ and $\mathbb{E}Y^2 = b < \infty$. Then*

$$\frac{1}{2}\mathbb{E}\log^+ Y \geq \frac{a^2 \log a - \frac{3}{2}a^2 + 2a - \frac{1}{2}}{2b}. \tag{2.2}$$

Proof. For $0 \leq t < a$,

$$a - t \leq \mathbb{E}\big[Y\, \mathbf{1}_{\{Y>t\}}\big] \leq b^{1/2}\mathbb{P}(Y > t)^{1/2}.$$

The first inequality follows by bounding Y by t on $\{Y \leq t\}$, and the second is Cauchy–Schwarz. Consequently,

$$\mathbb{P}(Y > t) \geq \frac{(a-t)^2}{b}. \tag{2.3}$$

The identity $\log^+ y = \int_1^\infty \mathbf{1}_{\{y>t\}}\, dt/t$ and Tonelli's theorem now give

$$\frac{1}{2}\mathbb{E}\log^+ Y \geq \frac{1}{2b}\int_1^a \frac{(a-t)^2}{t}\, dt = \frac{a^2 \log a - \frac{3}{2}a^2 + 2a - \frac{1}{2}}{2b}.$$

□

The hypotheses involve only moments, so the estimate applies without a density assumption. To use it for F_k, the essential point is that its fourth moment is bounded by an absolute multiple of the square of its second moment, despite the relations among its characters.

Lemma 2.2. *If X has normalized Haar distribution on $\mathbb{T}^d$, then*

$$\mathbb{E}|F_k(X)|^2 = k, \qquad Q_k := \mathbb{E}|F_k(X)|^4 = \begin{cases} 2k^2 - k, & k \text{ odd}, \\ 3k^2 - 3k, & k \text{ even}. \end{cases} \tag{2.4}$$

In particular, $Q_k \leq 3k^2$.

Proof. For every $b \in \mathbb{Z}^d$, product integration gives character orthogonality in the form

$$\int_{\mathbb{T}^d} x^b\, d\mu(x) = \begin{cases} 1, & b = 0, \\ 0, & b \neq 0. \end{cases}$$

On the torus $\overline{x^a} = x^{-a}$, so the relevant expansions are

$$|F_k(x)|^2 = \sum_{i,j=0}^{k-1} x^{a_i - a_j}, \qquad |F_k(x)|^4 = \sum_{i,j,u,v=0}^{k-1} x^{a_i + a_j - a_u - a_v}.$$

The vectors a_j are distinct, and therefore only the k diagonal terms survive integration in the first expansion. In the second, the surviving terms are exactly the balanced exponent sums:

$$Q_k = \#\{(i,j,u,v) \in \{0,\dots,k-1\}^4 : a_i + a_j = a_u + a_v\}.$$

Applying the coordinate map in (1.1) sends $a_i + a_j$ to $\zeta_k^i + \zeta_k^j$; its injectivity gives the reverse implication as well. Thus no additive relation is lost or introduced. If $\mu_k = \{z \in \mathbb{C} : z^k = 1\}$, it follows that

$$Q_k = \#\{(z_1,z_2,z_3,z_4) \in \mu_k^4 : z_1 + z_2 = z_3 + z_4\}. \tag{2.5}$$

For a nonzero sum $s = z + w$ with $|z| = |w| = 1$, an endpoint z must satisfy

$$\operatorname{Re}(z\overline{s}) = \frac{|s|^2}{2}.$$

This line meets the unit circle in at most two points. Since the endpoints are exchanged by $z \mapsto s - z$, the sum determines the unordered pair $\{z, w\}$, including the coincident-endpoint case. The only remaining possibility is $s = 0$, for which $w = -z$.

When k is odd, no pair of roots is antipodal. The k diagonal pairs give k contributions to (2.5). Each of the $k(k-1)/2$ unequal unordered pairs has two orderings on each side of the equality, and therefore gives four contributions. Hence $Q_k = k + 2k(k-1) = 2k^2 - k$.

When k is even, the zero sum has k ordered representations, one for each choice of the first root, and contributes k^2. The diagonal sums contribute k. Removing these k antipodal pairs and k diagonal pairs from the k^2 ordered pairs leaves $k^2 - 2k$. Each of their nonzero sums has two ordered representations. Thus

$$Q_k = k^2 + k + 2(k^2 - 2k) = 3k^2 - 3k.$$

For even k, negation permutes μ_k, so this balanced count is also the zero-sum four-step count in [3, p. 606]. For odd k, the balanced form (2.5) is essential. □

3 Proof and the growth conjecture

Proof of Theorem 1.1. By (2.1), the required measure is $\frac{1}{2}\mathbb{E}\log^+ Y$ for $Y = |F_k(X)|^2$. Lemma 2.2 gives $\mathbb{E}Y = k$ and $\mathbb{E}Y^2 = Q_k \le 3k^2$. We first obtain logarithmic growth from the single threshold $t = \sqrt{k}$ in (2.3):

$$\mathbb{P}(Y > \sqrt{k}) \ge \frac{(k - \sqrt{k})^2}{3k^2} = \frac{(1 - k^{-1/2})^2}{3}.$$

On this event, $\log^+ Y \ge \frac{1}{2}\log k$, so

$$\mathrm{m}(C_k) \ge \frac{(1 - k^{-1/2})^2}{12}\log k \ge c_0 \log k.$$

This proves the lower bound in (1.4).

To obtain the sharper lower estimate in (1.3), apply the integrated bound of Lemma 2.1:

$$\mathrm{m}(C_k) \ge \frac{k^2 \log k - \frac{3}{2}k^2 + 2k - \frac{1}{2}}{2Q_k}. \tag{3.1}$$

Its numerator equals $\int_1^k (k-t)^2\, dt/t > 0$. Replacing Q_k by its upper bound $3k^2$ proves the lower estimate in (1.3).

Finally, P_k has $k+1$ distinct monomials with coefficient one. Jensen's inequality and character orthogonality give

$$\mathrm{m}(P_k) = \frac{1}{2}\int_{\mathbb{T}^{d+1}} \log|P_k|^2\, d\mu \le \frac{1}{2}\log\int_{\mathbb{T}^{d+1}} |P_k|^2\, d\mu = \frac{1}{2}\log(k+1).$$

This reproduces the upper bound of [1, Proposition 6.1.2] and completes the proof. □

Conjecture 9.2.1 of [1] asks for a logarithmic lower bound when the parameter k is selected by an arithmetic condition, allowing an exceptional set of odd integers of natural density zero. To make this substitution, let $\kappa(n)$ be the least positive integer k for which $kn+1$ is prime. Such a k exists by Dirichlet's theorem. For odd $n>1$, the integer $n+1$ is even and greater than two, so $\kappa(n)\ge 2$.

Corollary 3.1. *For every odd integer $n>1$,*

$$\mathrm{m}(C_{\kappa(n)}) \ge c_0 \log\kappa(n).$$

In particular, Conjecture 9.2.1 of [1] holds for every such n, strengthening its almost-always conclusion.

Proof. Apply (1.4) with $k=\kappa(n)$. □

Together with the upper bound, the same result gives the two-sided order requested in the earlier thesis formulation [2, Conjecture 6.8.4]. The further comparison with Gaussian-period Mahler measures in [1, Conjecture 9.2.2] requires a varying-conductor estimate that is not supplied by the present argument.

4 The information used by the proof

The following examples explain the roles of the extra variable and the uniform fourth-moment bound in this argument. For $k=2$, one has $F_2(x)=x+x^{-1}$ and $\mathrm{m}(F_2)=0$, whereas the theorem gives $\mathrm{m}(x_0+F_2)>0$. Jensen's reduction in (2.1) thus cannot be replaced by an assertion about the ordinary Mahler measure of F_k.

The second moment alone is also insufficient. For

$$D_k(x) = 1+x+\cdots+x^{k-1},$$

orthogonality gives $\int_{\mathbb{T}}|D_k|^2\,d\mu = k$, but the geometric-sum formula implies, almost everywhere,

$$|D_k(e^{i\theta})| \le \frac{1}{|\sin(\theta/2)|}.$$

Consequently,

$$\mathrm{m}(z+D_k) = \int_{\mathbb{T}} \log^+|D_k|\,d\mu \le -\frac{1}{2\pi}\int_0^{2\pi}\log|\sin(\theta/2)|\,d\theta = \log 2.$$

Its fourth moment is

$$\int_{\mathbb{T}}|D_k|^4\,d\mu = k^2 + 2\sum_{r=1}^{k-1} r^2 = \frac{2k^3+k}{3},$$

which fails the uniform quadratic bound available for F_k.

The coarse moment conditions $\mathbb{E}Y = k$ and $\mathbb{E}Y^2 \le 3k^2$ do not determine the sharp leading asymptotic. Indeed, the abstract random variable that equals $3k$ with probability $1/3$ and zero otherwise satisfies $\mathbb{E}Y = k$ and $\mathbb{E}Y^2 = 3k^2$, while

$$\frac{1}{2}\mathbb{E}\log^+ Y = \frac{1}{6}\log(3k).$$

Thus the inequalities $\mathbb{E}Y = k$ and $\mathbb{E}Y^2 \le 3k^2$, by themselves, cannot force a leading coefficient greater than $1/6$. This example is not asserted to arise from a cyclovariety; it identifies the limit of the moment information used in the proof. A general asymptotic with leading term $\frac{1}{2}\log k$ would require additional control of the distribution of F_k.

AI assistance

AI assisted the formulation, literature comparison, proof development, and preparation of this manuscript within the Mathematics Knowledge and Connections Program.

This assistance extended to drafting and revising the abstract and Sections 1–4, developing and checking the mathematical arguments and proofs, and identifying, comparing and preparing the references. The author takes responsibility for the manuscript's content and citations.

References

[1] G. Cornelissen, D. Hokken and B. Ringeling, *The asymptotic Mahler measure of Gaussian periods*, Proceedings of the London Mathematical Society **133** (2026), no. 2, e70198. doi:10.1112/plms.70198. Open preprint: arXiv:2507.09303v3.

[2] D. P. T. Hokken, *Randomness and symmetry of polynomials*, Ph.D. thesis, Utrecht University, 2026. doi:10.33540/3576.

[3] G. Labelle and A. Lacasse, *Closed paths whose steps are roots of unity*, Discrete Mathematics & Theoretical Computer Science Proceedings, FPSAC 2011, AO (2011), 599–610. doi:10.46298/dmtcs.2937.

[4] R. Salem and A. Zygmund, *Some properties of trigonometric series whose terms have random signs*, Acta Mathematica **91** (1954), 245–301. doi:10.1007/BF02393433.

[5] C. J. Smyth, *On measures of polynomials in several variables*, Bulletin of the Australian Mathematical Society **23** (1981), no. 1, 49–63. doi:10.1017/S0004972700006894.